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Ontologies for Sustainability: Theoretical Challenges

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ABSTRACT

The lack of sustainable practices in key sectors of human societies is a global concern, posing threats for the whole planet. This paper provides the first review of ontologies for sustainability efforts, outlining their roles in assessing and advancing such efforts. Section 2 outlines three major challenges associated with sustainability: (1) the polysemy of the term; (2) the relationship between sustainability and sustainable development; and (3) the complexity underlying sustainability. Section 3 reviews the main accomplishments achieved by ontologies to meet these theoretical challenges. Special attention is devoted to ontologies belonging to the Open Biological and Biomedical Ontology (OBO)-Foundry community, due to the pivotal role these play in the field of applied ontology. A List of Sustainability Ontologies documents the assessed resources. Finally, Section 4 advances a potential trajectory to define an OBO Foundry-compliant family of sustainability ontologies and introduces a Sustainability Core Ontology (SCO) for consistently structuring this family.

1 | Introduction

Human activities and behaviours exhibited by organizations, societies, and individuals largely contribute to global issues such as climate change, land and ocean degradation, as well as biodiversity loss (IPBES 2018, 2019; UNEP 2021, 2022). Evidence abounds that these issues critically intensify year after year (Fanzo and Davis 2021; FAO et al. 2021; IPBES 2018, 2019; UNEP 2021, 2022; World Bank 2022). Imperative, viable, and long-term transformative actions intending to establish a more sustainable world are required.

Sustainability efforts (i.e., efforts to establish more sustainable practices) have an established history (Bolis et al. 2014; Du Pisani 2006; Purvis et al. 2019). Nevertheless, the semantic and, more generally, linguistic aspects of such efforts are still largely debated (Hariram et al. 2023; Piso et al. 2016; Salas-Zapata and Ortiz-Muñoz 2019). Selecting and sharing terms to be employed in representing different stakeholders' perspectives is a necessary step to co-design effective transformative solutions.

Several initiatives have aimed to unearth the implicit meaning of "sustainability" and its cognate expressions. Lists of issues, indicators' catalogues, portals, thesauri, and databases count among them. Efforts, however, have primarily focused on tracking sustainability features in specific domains, without considering the challenging nature of cross-domain conversations. Therefore, to date, there still subsist severe semantic barriers that impede coordination of sustainability initiatives across domains and communities.

Ontologies have provided valuable tools to secure semantic links and interoperability in complex communication settings, such as the biomedical field, whose terminology is now streamlined through the Open Biological and Biomedical Ontology (OBO)-Foundry (Smith et al. 2007). To date, no analogous exists—yet—for the sustainability domains. To our knowledge, no study has either assessed available ontologies regarding sustainability. This paper offers a first systematic review, exploring ontologies' role as conceptual and semantic guides for facilitating cross-domain understandings of sustainability and for coordinating

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related transdisciplinary research and interventions. Further, it outlines an initial roadmap toward establishing an initiative in line with the OBO-Foundry.

In Section 2, we offer a brief overview of ontologies, focusing on those pertaining to applied ontology—the field of study that encompasses research efforts at the intersection of philosophy and computer science—remarking on their suitability for sustainability. In Section 3, we review the three main theoretical challenges associated, in contemporary literature, with sustainability efforts: (1) the polysemy of the term, (2) the relationship between sustainability and sustainable development, and (3) the complexity underlying sustainability. In Section 4, we assess currently existing ontological resources, linking them to these challenges. We perform a literature review and we explore major available ontology portals. Our analysis devotes particular attention to ontologies belonging to the OBO-Foundry community, due to the pivotal role these play in the domain of applied ontology. We provide a List of Sustainability Ontologies (Annex 1), collecting the resources we assessed. In Section 5, we propose a potential trajectory to further develop an OBO Foundry-compliant family of sustainability ontologies and we develop a Sustainability Core Ontology (SCO) (Annex 2) to consistently structure this family.

2 | What Is an Ontology?

To start, we shall provide a brief overview of ontologies, highlighting their conceptual and semantic guidance for cross-domain sustainability efforts as well as for coordinating transdisciplinary research and intervention.

2.1 | Ontologies

The term "ontology" developed within philosophical quarters, being employed to designate disputes on the nature and structure of the entities in a given domain of discourse (e.g., whether rocks are anything over and above the particles that compose them, or whether people have a dual nature of matter and soul). According to a widely shared division of the field of study, "formal ontology" focuses on the elucidation of general features of entities, such as existence, identity, or parthood; instead, "material ontology" and "domain ontologies" deal with fundamental aspects of reality in relation to specific domains of interest, such as material objects, biomedical entities, or digital objects (Arp et al. 2015; Hennig 2008; Smith 2003; Varzi 1999, 2005).

During the last few decades, ontologies have gained prominence in informatics and computer science (Arp et al. 2015; Smith 2003), within which they now comprise the field of *applied ontology*. Within this research context, "ontology" stands for a theory of knowledge representation, typically arranged in a hierarchically organized corpus of terms supported by a canonical syntax, relational axioms, and a stock of agreed upon definitions and universal IDs (Dooley et al. 2018; Griffiths et al. 2017; Smith 2003, 158). An ontology, intended in this specific sense, establishes a lexical or taxonomic framework, ensuring a common semantic standard for accessing views and related data, promoting their integrability and interoperability among its users. Ontologies for informatics and computer science are generally developed in open access, fostering their reusability, shareability, and joint maintenance and validation across stakeholders and fields of study. In this paper, we use the term "ontology" as employed within the field of applied ontology.

2.2 | The OBO-Foundry

The Open Biological and Biomedical Ontologies (OBO) Project plays a prominent role in the field of applied ontology. The project started in the early 2000s to strengthen data interoperability within the domain of life sciences (Jackson et al. 2021; Smith et al. 2007). It advanced a set of shared principles regarding ontology design, including that ontologies should be developed as open, orthogonal to each other, and syntactically ordered resources that hold a "common system of identifiers" (Jackson et al. 2021; Smith et al. 2007, 3). The original set of principles was later expanded by the OBO-Foundry, requiring ontologies to be designed through a collaborative effort, employ common and no-misunderstanding relations, provide a system of user feedback and version tracking, and address a specific domain of interest (Smith et al. 2007). Recently, OBO principles undertook further elaboration and refinement (Jackson et al. 2021).

To date, OBO Foundry includes 181 ontologies, reaching beyond the initial life sciences focus (http://obofoundry.org/; 11/02/2023). Basic Formal Ontology (BFO) plays a pivotal role among OBO Foundry ontologies. All resources, which are part of this foundry, employ BFO as the upper-level ontology, extending their representations from BFO representation (Arp et al. 2015). In fact, BFO offers a set of shared general terms, providing a shared foundation for different domain-specific vocabularies (Arp et al. 2015). Thus, BFO ensures compliant ontologies to represent domain-related entities "in a consistent and coherently structured fashion" (Arp et al. 2015, 87). Providing a common top-level representation, BFO further fosters integrability and interoperability across the foundry resources (Arp et al. 2015). BFO is hierarchically small, encompassing 36 terms (Arp et al. 2015; Smith 2012; ISO/IEC 21838-2:2021 2021). The last released version is the BFO 2020, which is a recognized top-level ontology (TLO) conforming to the ISO/IEC 21838-1 TLO standard (ISO/IEC 21838-2:2021 2021).

2.3 | Ontologies for Sustainability

Establishing explicit, structured, and logically sound vocabularies, ontologies can contribute to effectively clarifying, organizing, and linking proposals and data on sustainability (Smith 2003). As such, ontologies may foster the development of consistent and viable transdisciplinary perspectives, corroborated by sound and fair data, which can be integrated across stakeholders and communities, thereby facilitating related research and intervention efforts in a reliable and non-misinterpreted manner. Further, due to the continuous and open curation process to which ontologies are subjected, these resources can equally and constantly guarantee a coherent and actionable tracking of possible progress in the sustainability narrative.

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3 | Sustainability: Theoretical Challenges

We explored sustainability literature for a period of 2 years (2022–2024), focusing primarily on publications engaging directly with "sustainability" and "sustainable development," and cognate expressions. Authors of the surveyed works, more or less explicitly, made reference to three major theoretical challenges when it comes to defining and determining what sustainability is: (1) the polysemy of the term; (2) the relationship between sustainability and sustainable development; and (3) the complexity underlying sustainability issues. To our knowledge, no author has accounted for these challenges jointly, nor has prioritized an analysis of sustainability that is grounded on these challenges. We therefore devote this section to analyze them, as they appear essential to effectively coordinate cross-domain efforts as well as to encourage consistent transdisciplinary coordination of transformative actions for sustainability.

3.1 | Challenge 1: The Polysemy of Sustainability

The first theoretical challenge concerns the polysemy of the term "sustainability," that is, the fact that two or more related meanings are associated with the same *word form* (Sennet 2021; Vicente and Falkum 2017). Authors dealt with sustainability across several disparate fields of study and practice during a long time span. They have backed multifarious perspectives, associating multiple meanings with the term. To exemplify, we present some major perspectives that emerged during the last decades.

In the post-Second World War context, several authors criticized the economic vision of the time by showing an improper use of the terms "development" and "economic growth" (see, for instance, Goldsmith (in *Blueprint for Survival* 1972) and Schumacher (in *Small is Beautiful* 1973)). These terms—the critique alleged—were used interchangeably (Purvis et al. 2019), yet a development conceived as infinite economic growth is not sustainable on a finite planet. The *Limits to Growth* by Meadows et al. (1972), commissioned by the Club of Rome, emerged as the manifesto of this perspective, demanding a "sustainable world system" (Purvis et al. 2019, 682), where a "global equilibrium" will ensure that each person can fulfill their proper material needs, with an "equal opportunity" to achieve their "individual human potential" (Meadows et al. 1972, 24).

In 1987, the United Nations World Commission on Environment and Development (UN WCED) released the report "Our Common Future," also known as the "Brundtland Report" (Purvis et al. 2019; Redclift 2005). This document coined one of the most famous and debated definitions of sustainable development: "the development which meets the needs of current generations without compromising the ability of future generations to meet their own needs" (World Commission on Environment and Development 1987, 16). The definition underlines the two major features at the basis of the "Our Common Future" perspective. The first concerns *who* sustainability matters to (present and future generations), and the second one relates to *what* matters (intergenerational ability to satisfy own needs) (DesRoches 2021). This perspective brought the sustainability debate into the view of a wider international public (Purvis et al. 2019; Redclift 2005).

After the publication of the Brundtland Report, the "Socio-Scientific Approach" emerged in the economics fields of study as another influential perspective on sustainability. Here, the debate focused on maintaining the available capital—stocks and flows of resources, human attributes, human capacities, and means—of an economy over time to sustain development (DesRoches 2019; Meadows 1998). To maintain capital over time, its "saving rate" must be higher or equal to that of "deprecation" (Cabeza Gutés 1996, 147; DesRoches 2019). Two main opposite positions stand on capital maintenance, mainly focusing on the substitutability among different kinds of capital (social, human, built/manufactured, and natural) (Cabeza Gutés 1996; DesRoches 2019; Pelenc et al. 2015). Of special importance was the substitutability between the built and natural capital. According to Weak Sustainability, the two kinds of capital should be recognized as fully exchangeable: a loss of natural capital from one generation to the other is acceptable if compensated by an increase in the built one (Ayres 1998; DesRoches 2019; Pelenc and Ballet 2015); it only matters that the total aggregate amount of capital does not diminish across generations, as technological innovation will ensure overcoming possible environmental implications (Bolis et al. 2014; Pelenc et al. 2015). In contrast, supporters of Strong Sustainability argue for a not-absolute substitutivity between built and natural capitals (Ayres 1998; Bolis et al. 2014; DesRoches 2019). The built capital depends on the natural one for its production (Pelenc and Ballet 2015). The destruction of the built capital is reversible, while that of the natural capital is not (Bolis et al. 2014; Pelenc et al. 2015). The natural capital is multifunctional and multidimensional (Pelenc and Ballet 2015). It encompasses some critical features (Critical Natural Capital) that uniquely contribute to human well-being and that no other kind of capital can substitute (DesRoches 2019; Pelenc and Ballet 2015). Thus, Strong Sustainability envisages natural capital and built capital as complementary, claiming a simultaneous—not in aggregate—maintenance of them through time (DesRoches 2019; Pelenc and Ballet 2015; Pelenc et al. 2015).

At the end of the 20th century, authors started to characterize sustainability in terms of "dimensions" or "pillars." The three main ones being: the environmental, the social, and the economic pillar (Lehtonen 2004, 200; Purvis et al. 2019, 681). Pillars are generally represented through Venn diagrams or embedded circle-based illustrations (Lozano 2008). In the business spheres, the "Pillars" perspective took the shape of the so-called "Triple Bottom Line (TBL)" theory (Elkington 1997). In *Cannibals with Forks*, Elkington (1997) proposed to complement the traditional financial "Bottom line" with bottom lines addressing social and environmental features (Elkington 1997; Purvis et al. 2019). To date, the "Pillars" perspective is extensively diffused both in public and private discourses.

More recently, sustainability has been addressed through a series of interrelated goals. This goal-centered perspective gained international relevance with the adoption of the Millennium Development Goals (MDGs) by UN member nations at the beginning of the new millennium. MDGs focused on poverty, hunger, gender inequalities, education, and health conditions (UN Assembly 2001). The MDGs perspective rests on eight goals, linked to 21 targets, and assessed through 60 indicators (Liverman 2018). In 2015, the Sustainable Development Goals

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(SDGs) replaced MDGs. SDGs aim to "end poverty, protect the planet, and ensure prosperity for all," promoting actions to 2030 in the critical areas of People, Planet, Prosperity, Peace, and Partnership (United Nations 2015). SDGs count 17 goals, specified by 169 targets, and monitored through 231 indicators (Estoque 2020; United Nations 2015). Despite being largely debated, SDGs are the principal international reference frame for sustainability we look at to date.

Sustainability polysemy presents a *double nature*. It is simultaneously a diachronic and a synchronic phenomenon. Sustainability polysemy occurs *through time* (Diachronicity). Many perspectives on sustainability have arisen over many centuries. Sustainability polysemy also occurs *at one specific time* (Synchronicity). At a specific historical moment, different positions stand over the same perspective, for instance, the Weak and Strong Sustainability for the Socio-Scientific Approach. Further, synchronicity also relates to the specific stances of authors. Disparate and sometimes even conflictual stances about the same theme may crystallise in peculiar points of view, leading to a multiplicity of different nuances of meanings over the same perspective. The historical moment and the related issues of the time as well as the authors' respective backgrounds, fields of research, values, and aims, stand as the basis of this double nature of sustainability polysemy.

Even if each perspective on sustainability is clearly identifiable, circumscribing them unambiguously is arduous. Some perspectives may derive from previous ones. Some features pertaining to a perspective may also be addressed by another one. Even if a new perspective arises, the previous ones may persist in the debate. Authors may change (partially or radically) their ideas, contributing to the development of new perspectives or improving already existing ones. Consequently, sustainability polysemy occurs as an *enchained continuum* of meanings *through* and *in* time.

Sustainability polysemy presents a *double shape*, too: authors not only attribute significance to the term "sustainability" by defining it, but also through the usage of this term. In this regard, Salas-Zapata and Ortiz-Muñoz (2019, 3–5) outline four different understandings of "sustainability:"

- A set of criteria: the "social-environmental criteria or qualities" guiding human actions in relation to a reference system.
- A vision: the goals that human actions point to in a reference system
- An object: an "entity that exists, that can be represented, studied or intervened," for example, a property or the behaviour of some reference system.
- An approach: the assessment of some "variables of a reference system."

These use-driven meanings pertain to the field of academic research. Due to the large employment of the term also in other contexts, additional use-driven meanings may exist.

3.2 | Challenge 2: The Relationship Between Sustainability and Sustainable Development

The second theoretical challenge refers to the controversial relationship between sustainability and sustainable development. Both in the public and private spheres, authors deal with sustainability and sustainable development in the same narrative, showing their strict relationship. Nevertheless, the nature of this relationship largely varies. Simplifying a bit for the purposes of organizing the literature, we can say that authors have primarily conceived sustainability and sustainable development in one of three ways: as equivalent or synonymous (Lozano 2008; Salas-Zapata et al. 2017); as independent of each other; or as interconnected and dependent on one another (Sartori et al. 2014). In most cases, however, authors do not provide straight definitions of sustainability and sustainable development (or deliberate uses of the terms). The underpinned meanings are only implicitly revealed in the proposed narratives, thereby leading to the same situation regarding the endorsed type of relationship (Salas-Zapata et al. 2017). This largely contributes to shaping the controversial picture of the sustainability–sustainable development relationship.

The term "sustainable development" is also polysemous (Lozano 2008; Robinson 2004; Sartori et al. 2014). The plurality of meanings, underpinned by the term "sustainable development" and that of "sustainability," appears to be, at least partially, overlapping in the literature. Some authors associate a particular significance to the term "sustainability" that others relate to that of "sustainable development," and vice versa (Sartori et al. 2014). This shift of meanings may occur only to some extent. An author may associate some aspects of meaning to the term "sustainability" that another one may refer to the term "sustainable development." The two authors may agree upon the remaining aspects or have different views. Such an overlapping polysemy complicates even further the understanding of sustainability and sustainable development as well as their existing relationship. Equivalence, independency, and dependency between sustainability and sustainable development turn up intrinsically interconnected to and influenced by this multiplicity of converging and overlapping meanings.

3.3 | Challenge 3: The Sustainability-Underpinned Complexity

The third theoretical challenge regards the grounding of sustainability in complexity. Such grounding on complexity relates to the "value judgement" that should accompany the multifarious meanings (and types of relationships) pertaining to sustainability and sustainable development (DesRoches 2021). In fact, authors generally agree on the "thickness" of sustainability and sustainable development. Thickness identifies those terms as underpinning both non-evaluative (descriptive) and evaluative features (DesRoches 2021). Descriptive features "help to give us a specific idea" of sustainability and sustainable development (Kirchin 2013, 2). The evaluative ones highlight sustainability "to be essentially keyed into approval and disapproval" (Kirchin 2013, 1). Expressing a value judgment is fundamental as sustainability and sustainable development unquestionably concern a multiplicity of human and non-human stakeholders, dealing with the related complexity of their specific situations

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and with the complexity of integrating their perspectives. Thus, sustainability and sustainable development are essentially grounded on complexity.

In the sustainability literature, emphasis on complexity generally emerges in relation to resilience, vulnerability, and adaptability of the Socio-Ecological systems (SESs), also called Human-Environmental systems. The appellation of SESs designs complex systems where humans are involved, accentuating the relation(s) between humans and the other system components (Folke et al. 2005; Haines-Young and Potschin 2012; Rutting et al. 2022). Nevertheless, the discussion on complexity covers even further themes.

Complexity is generally recognized as an intrinsic feature of our world (Heylighen et al. 2006; Mamédio and Meyer 2020), despite the lack of an agreement on its definition and of a "science of complexity" (Mitchell 2009, 94, 95). Disparate formal and informal characterizations exist, depending on the field of research (Mitchell 2009). A point of convergence of these characterizations may be identified in complex systems, with biological systems often featured as the "ideal" ones (Latora and Marchiori 2004, 1). Even though a shared definition of a complex system does not exist across fields of study, drawing on the works of Latora and Marchiori (2004), Mitchell (2009), Ramalingam et al. (2008), Sayama (2015) and Serrat (2017), we may characterize it for present purposes as follows: *a complex system is a system in which multiple interdependent components interact dynamically and non-linearly, causing the emergence of a collective nontrivial and self-organized behavior, across scales and levels of analysis, without any internal/external centralized controls.*

The first fundamental aspect which emerges from this definition is that a (complex) system is always interpreted in relation to a surrounding environment, from which the system is separated and identified by boundaries (Heylighen et al. 2006; Landgrebe and Smith 2023). Systems are connected among themselves and with their environment through input and output streams (material, energy, and information) (Heylighen et al. 2006). Input and output streams concatenate different systems into a network (Heylighen et al. 2006). If the network "functions in a sufficiently coherent manner," it shows up as a "supersystem," where initial systems present as non-independent subsystems of that new supersystem (Heylighen et al. 2006, 7).

To interpret the subsystem-supersystem relationship, we need to refer to scales and levels. A scale represents "the spatial, temporal, quantitative, or analytical dimensions used to measure and study any phenomenon" (a complex system), whereas levels "the units of analysis" are located along those scales (Gibson et al. 2000, 218, 219). Complex systems architectures, behaviours, and properties are conceived in terms of system scales and levels.

Further, the proposed definition highlights the two key features of complex systems: emergence and self-organization. Emergence points out that the complex system behavior and properties (either rules) at a macroscopic level are non-trivially envisaged from those at the more microscopic ones (Heylighen et al. 2006; Sayama 2015). Instead, self-organization highlights that system components spontaneously arrange, globally generating macroscopic and microscopic structures and behaviors, without any internal or external controls (Mitchell 2009; Sayama 2015).

Our definition also emphasizes the relevance of interconnectedness and interdependence of the complex system components among each other and their environment. Interconnectedness and interdependence occur across scales and levels as well as across different complex systems (Ramalingam et al. 2008). The "complex changing behaviour" of a complex system at a certain level essentially reflects the interactions among its interdependent components at that level (Mitchell 2009, 15; Ramalingam et al. 2008). Authors often employ the term dynamics (or dynamical/dynamics systems) to describe this changing nature (Allen and Prosperi 2016; Mitchell 2009). Such dynamics appear strictly nonlinear across complex systems with no proportionality elapsing between causes and effects (Heylighen et al. 2006; Ramalingam et al. 2008; Serrat 2017). Sensitivity to initial conditions equally shapes dynamics within complex systems. In two very close complex systems (in terms of components and scales), small differences in the initial system state may lead to two unpredictable largely different scenarios at a later stage (butterfly effect) (Heylighen et al. 2006; Ramalingam et al. 2008). Possible positive and negative feedback may also occur, contributing to the nonlinear complex system behaviours (Heylighen et al. 2006; Ramalingam et al. 2008).

Agency is a further fundamental feature. Normally, authors conceive complex systems as multi-agent, where several autonomous and distinct components (called agents) interact among themselves and with their environment (Heylighen et al. 2006). Agents sense "their internal state" and their surroundings (Heylighen et al. 2006; Ramalingam et al. 2008, 42). They may display "internal information processing and decision-making capabilities," learn, and save knowledge (Ramalingam et al. 2008, 42, 43). Agents display adaptiveness (Ramalingam et al. 2008). They coevolve, adapting to system and environment component changes (Holland 2006; Ramalingam et al. 2008). Sometimes, authors employ the term "complex adaptive systems" to identify those systems that are strongly characterized by adaptiveness, such as the human immune system or food systems (Mitchell 2009).

One last essential feature is worth stressing: the perspectives of *all* stakeholders within a complex system matter (to some extent) regarding sustainability. Each living component of the Earth's complex systems owns its own ensemble of ideal conditions for living, sustaining, and lasting. In this sense, each living component bears a *sustainability perspective*. While not all authors would agree on the relative importance of individual perspectives, it seems fairly agreed upon in the literature that multiple perspectives, beyond those of (some) humans, must be taken into account (see more-than-human environmental ethics works, such as the classic Callicott (1984)). Furthermore, it appears required to give substance to these perspectives, even if setting them from a human point of view will lead to simply conceiving proxies of the real perspectives.

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4 | Ontologies and Sustainability

We posit that a realistic and robust ontological representation of sustainability requires dealing consistently with each of the challenges that we just presented. In this section, we move to evaluate available ontological resources in terms of these challenges.

4.1 | Methods

We proceeded in two parallel paths to look for ontologies that address sustainability and sustainable development: we performed a literature review (Path 1) and we queried for specific keywords in the major available ontology portals (Path 2), (period: 06/2022–02/2024; location: Italy and USA).

Path 1. We explored Google Scholar and PubMed as browsers to search for scientific literature, employing the following 12 combinations of keywords:

- Sustainability—Ontology;
- Sustainable development—Ontology;
- Sustainable Development—Sustainability—Ontology;
- OBO Foundry—Sustainability;
- OBO Foundry—Sustainable Development;
- SDGs—Ontology;
- MDGs—Ontology;
- Sustainability—Complexity—Ontology;
- Sustainability—Polysemy—Ontology;
- Sustainable Development—Complexity—Ontology;
- Sustainable Development—Polysemy—Ontology;
- Sustainabilit—Sustainable Development— Complexity—Polysemy— Ontology;

Keyword choice was based on the assessment review of sustainability literature we performed in Section 3.

In case of large amounts of outcomes, we limited our assessment to the first 100 results of each query. We manually selected relevant articles among matching results and their related bibliographies according to themes characterizing the sustainability debate, explored in Section 3.

Path 2. Ontologies are generally openly accessible through online portals, such as OntoBee linked data server browser (Ong et al. 2017; Xiang et al. 2011), EMBL—EBI Ontology Lookup Service (OLS) (Jupp et al. 2015), NCBO BioPortal (Whetzel et al. 2011), and AgroPortal (Jonquet et al. 2018). To date (06/26/2023), these portals host 262 (OntoBee), 280 (OLS), 1054 (BioPortal), and 150 (AgroPortal) ontologies. We explored these four resources. We scrolled portal indexes and ontology descriptions. Then, we queried for the following keywords, assessing results and the related ontological taxonomies:

- Sustainability;
- Sustainable Development;
- Weak Sustainability;
- Strong Sustainability;
- Capital (natural, social, human financial, built);
- Socio—Scientific Approach;
- Resilience Paradigm;
- Limits to Growth;
- Brundtland Report;
- Our Common Future;
- Triple Bottom Line;
- Bottom Line;
- Millennium Development Goals;
- MDGs;
- Sustainable Development Goals;
- SDGs;
- Need;
- Capability(ies);
- Capability(ies) Approach;
- Well(-)being;
- Polysemy;
- Ambiguity;
- Diachronicity;
- Synchronicity;
- Meaning;
- Complexity;
- Complex System;
- System;
- Adaptive System;
- Dynamic/Dynamical System;
- Dynamics;
- Networks;
- Social-ecological System;
- Ecosystem;
- Reference System;

In Agroportal, we also employed the plurals of the keywords, as we noticed that many ontologies included plural nouns. Also in this case, we chose keywords on the basis of the reviewing assessment of sustainability literature we performed in Section 2 and, in case of large amounts of findings, we limited our term assessment to the first 100 items of each query. We manually selected relevant terms and related ontologies among matching

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results according to themes characterizing the sustainability debate, assessed in Section 3.

4.2 | Cataloguing

We documented retrieved ontologies and terms in a List of Sustainability Ontologies (Annex 1). Annex 1 covers two sublists, List A and List B. List A presents ontologies retrieved both through Path 1 and Path 2. For each ontology, List A outlines the following information: the ontology name, the acronym, a reference, and the modality by which we gathered it (literature review, portals' exploration, or both modalities). List B presents terms collected through Path 2. For each term, List B outlines the name, the reference IRI, and the related ontology (acronym). Broadly, Annex 1 documents 574 ontologies: 282 retrieved through Path 1, 251 from Path 2, and 41 through both paths. Annex 1 also documents 1022 terms retrieved through the exploration of the ontology portals. See the annex guidelines for more details.

This effort aims to systematize information collected in this assessment and make them available to interested stakeholders. We are aware this list is not over-comprehensive of all existing ontologies or terms related to sustainability and sustainable development. As far as we know, however, no previous study has attempted to enumerate together all available ontological contents related to sustainability. Thus, this resource aims to be a starting point.

4.3 | Ontology Review

Here, we briefly present the results of our assessment. We begin with ontologies addressing sustainability per se, to then consider also those ontologies that deal with sustainability with respect to a specific domain of interest. Along the road, we provide examples.

The sustainability ontology scenario is strongly varied. Few ontologies focus on sustainability and sustainable development per se. They almost all relate to the frame of the UN SDGs. To illustrate, we present two of them: the Sustainable Development Goals Interface Ontology (SDGIO) (UNEP 2016) and the Common Ontological Sustainable Development Meta-Model (Deliyska et al. 2020). SDGIO is the official UN interface ontology for SDGs (Smith and Jensen 2016; UNEP 2016). It is compliant with the OBO-Foundry and its practices. Instead, the Meta-Model proposes a structured set of ontologies, focusing specifically on sustainable development. It is the only one we retrieved that is not part of the UN frame.

The SDGIO design started in 2015, at the request of the Inter-Agency Expert Group on Sustainable Development Goals (IAEG-SDG) to the UN environment program (UNEP) and experts in the field (FAO 2016; UNEP 2016; Buttigieg et al. 2016). SDGIO represents SDGs and related entities, seeking to foster SDGs data interoperability by clarifying the meanings associated with related terms (Jensen 2016; Joshi et al. 2021). The embraced perspective on sustainability is precisely that of SDGs. V2018-08-10 is the currently available version. As part of the OBO Foundry, SDGIO embraces OBO principles and employs BFO as the upper-level ontology (Jensen 2016; Smith and Jensen 2016). SDGIO appears strongly designed for interfacing (Joshi et al. 2021, 12). It imports terminologies from several ontological resources, such as CHEBI, DOID, EnvO, GEMET, IAO, NCBITaxon, OBI, OGMS, PATO, PCO, and UBERON (Buttigieg et al. 2016; Bianchetti et al. 2018; Smith and Jensen 2016).

In 2018, a group of researchers from the University of Forestry (Bulgaria) started to develop a Meta-Model, aiming to systematize knowledge related to sustainable development (Deliyska et al. 2020). The fundamental adopted perspective appears to be that of the "three pillars." Sustainable development is conceived as "a process and a status of some system" (Deliyska et al. 2020, 59; Ivanova and Deliyska 2022, 2; Ivanova et al. 2018, 2; Ivanova et al. 2021, 2). The term "sustainability" is used as a synonym. Authors embrace a "system theory point of view" (Deliyska et al. 2020, 59; Ivanova and Deliyska 2022, 2; Ivanova et al. 2018, 2; Ivanova et al. 2021, 2). The Meta-Model comprises four ontologies: an upper-level ontology, named the Common Ontology of Sustainable Development (COSD) (Deliyska et al. 2020), and three domain ontologies, called Ontology of Sustainable Development in Economy (OSDE) (Ivanova et al. 2018), Ontology of Social Sustainable Development (OSSD) (Ivanova et al. 2021), and Ontology of Environmental Sustainable Development (OESD) (Ivanova and Deliyska 2022). The major hierarchy division in all these resources occurs between classification (subsumption) and mereology branches. The classification branch provides the *is_a/has_a* relations, and the mereology branch provides the *part-whole* ones. OBO Foundry includes no Meta-Model ontology. Nevertheless, COSD, OESD, and OSSD import classes, among the others, from SDGIO.

A considerably larger number of ontologies address sustainability in relation to some specific domains of knowledge. Domains show up in a disparate, hard-to-systematize fashion. The most represented are the urban, indicator/assessment, and company/business domains. We present the Food-Energy-Water Ontology (FEWsOnt) as an example (Babaie et al. 2019). This ontology strongly models sustainability from a complex system perspective (Babaie et al. 2019). Further, FEWsOnt does so in terms of three systems fundamental to human survival: food, energy, and water systems.

Researchers developed FEWsOnt in 2018. This resource represents the dynamic structures and interactions occurring across food, energy, and water complex systems. A vigorous awareness stands on the "nexus" and interdependence and influence among these three systems (Babaie et al. 2019, 808, 809, 811). FEWsOnt primarily addresses major complex system features, such as emergence, self-organization, adaptation, and co-evolution (Babaie et al. 2019). Authors developed this resource, aiming to support knowledge discovery through logical inferences and to foster communication among stakeholders in terms of sustainable development (Babaie et al. 2019). FEWsOnto conceives sustainability as an "emergent property of a desirably functioning complex global economic–environmental–social system," categorizing systems processes and properties in terms of "effects" on sustainability (Babaie et al. 2019, 810, 812, 813). The embraced perspective appears to be the one relying on the "three pillars." Particular attention is paid to climate change

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(Babaie et al. 2019). This ontology employs BFO as the upper-level ontology (Babaie et al. 2019).

Finally, the large majority of the retrieved ontologies only encompass some terms (generally just a few isolated ones) referring to sustainability and sustainable development without addressing them extensively. The latter occurs more often than the former.

4.4 | Discussion

We now briefly discuss the results of our ontology review, based on how they deal with the three theoretical challenges we outlined in Section 3. Along the road, we provide examples.

4.4.1 | Ontologies and Challenges 1 and 2

In terms of sustainability theoretical challenges, we can summarize that no one singular ontology wholly engages with nor emphasizes the challenging nature of sustainability. We could not find an ontology that consistently addresses sustainability polysemy. Different perspectives appear across the retrieved ontologies, with each ontology generally focusing on one singular perspective on sustainability. The "Our Common Future" perspective is represented the most, followed by those of SDGs and the "three pillars." Rarely, some ontologies engage with the perspectives of the TBL or the MDGs. Interestingly, we retrieved an ontology, called the Strong Sustainable Business Model Ontology (SSMO) (Upward and Jones 2016), which represents sustainability in the business domain specifically from the perspective of Strong Sustainability. A few resources, such as AGROVOC or GEMET, propose terms and definitions referable to more than one perspective, generally the "Our Common Future," SDGs, and MDGs perspectives. Sustainability polysemy does not appear addressed in either of those cases. Perspectives on sustainability are generally confused and there are no straightforward accounts of them. Sustainability polysemy is not even addressed considering different resources together. Similarly, no retrieved ontology dedicates attention to possible meanings underpinned by the uses of the term sustainability. The same considerations regard the types of relationship and the overlapping polysemy between sustainability and sustainable development.

The three ontological resources we mention above present clear examples of that.

- SDGIO aims to specifically address SDGs. As the ontology focus is strictly there, despite its pivotal role in shaping the semantics of sustainability within the UN frame, no insights into covering sustainability polysemy seem to emerge from the ontology structure or the related literature.
- The proponents of the Meta-Model mentioned the vagueness and ambiguity of the term "sustainability." They also contemplate several of the major related perspectives before finally representing sustainable development from the "three pillars" perspective, which they considered the most widespread one (Deliyska et al. 2020). They do not provide a clear and consistent account of the other perspectives which instead seem only encompassed in an incoherent and tangled manner. Sustainability is conceived as equivalent to sustainable development, which refers inconsistently to both a process and a system state.
- FEWsOnto makes no mention of the polysemous nature of these terms. Related literature shows FEWsOnt defining sustainability as a system property, underlining only the "three pillars" perspective. Nevertheless, we were not able to retrieve the ontology file. Thus, our considerations may be adjusted regarding this resource.

The only ontology which slightly differs from the others reviewed here is the Social Ontology on Sustainable Development Implementation at the Central University of Technology, Free State (CUT) (Awuzie et al. 2017). Authors point to designing this resource based on the different stakeholder perspectives on sustainability. To date, however, this ontology has either not been developed yet or, if so, not been released publicly.

4.4.2 | Ontologies and Challenge 3

The findings are quite varied with regard to sustainability-underpinned complexity. Generally, this third challenge appears disparately addressed, depending on which ontology is considered. Most of the retrieved ontologies just encompass a few terms relating to sustainability and complexity. Certain ontologies point to sustainability-underpinned complexity through its purpose and design or through the role that the represented domain plays in the sustainability narrative. This is the case of the Ontology for Sustainability Indicators Set (OSIS) (Ghahremanloo et al. 2012; Ghahremanlou 2014) or the Sustainability Science (SS) Ontology (Kumazawa et al. 2009). OSIS aims to uniquely represent current knowledge about indicators employed to measure the sustainability of complex systems (Ghahremanloo et al. 2012; Ghahremanlou et al. 2017, 2016). SS ontology seeks to support building and systematizing knowledge within the transdisciplinary field of sustainability science, principally from the perspective of what kind of problems sustainability science should solve and how (Kumazawa et al. 2009).

Other ontologies address complexity-related themes which are fundamental in the sustainability narrative, such as Life Ontology (LifO) (He 2018) and the Ecosystem Ontology (ECSO) (O'Brien and Schildhauer 2017). LifO represents "the life, different life tenets, and how various life elements are interacted in a dynamic and complex network pattern" (He 2018). It is grounded on the theory of OneNet, which states that "the whole process of the life of an organism is a single complex and dynamic network ('OneNet')" (He 2018). LifO employs BFO as upper-level ontology. ECSO describes "common terms and measurements used in the study of Ecosystems," focusing principally on those related to carbon-cycling (O'Brien and Schildhauer 2017; Jones et al. 2021). ECSO has been developed as part of DataONE, a "community-maintained" initiative (Jones et al. 2021). DataONE aims to foster open data access and interoperability among member repositories (Jones et al. 2021; The Ecosystems Ontology, n.d.).

Further, few ontologies address sustainability by directly grounding on complexity. Consider, for instance, the Service

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Ecosystem Ontology (Badawi et al. 2020) and the Ontology for Sustainable Urban Mobility System (Moskolaï et al. 2019). The Service Ecosystem Ontology relies on a systemic thinking based on "the Service Dominant-Logic (S-D Logic) perception of service ecosystems" (Badawi et al. 2020, 305). It addresses sustainability in relation to the complexity of actors' practices occurring during "their quest for survival in the surrounding dynamic environment" (Badawi et al. 2020, 305). Instead, the Ontology for Sustainable Urban Mobility System represents the sustainability of urban mobility, strongly taking into account the complex nature of urban systems.

Regarding the three ontological resources presented above, we can state the following considerations:

- SDGIO does not appear primarily committed to complexity. Nevertheless, for their own nature, SDGs point to that, proposing a multifaceted view of sustainability. SDGs Agenda outlines a large set of interrelated and potentially transformative complex themes (Fukuda-Parr 2016; Liverman 2018). Thus, SDGIO carries, in some way, this intrinsic propension to complexity itself.
- The Meta-Model interprets sustainable development from a system theory point of view. For that reason, Meta-Model resources refer to the systems-processes terminology which belongs to the Common Ontology of System (COS) (Deliyska et al. 2020). COS represents knowledge focused on systems and energy and matter transfer processes (Deliyska 2019). This effort points to addressing sustainable development in terms of complex systems.
- FEWsOnto shows up specifically for its strong commitment to complexity. FEWsOnt addresses sustainability based on and in relation to three major complex systems, fundamental for human survival.

4.4.3 | Sustainability Ontologies and Interoperability

Along with faintly engaging with the theoretical challenges to sustainability, we noted a general lack of integration and interoperability among assessed ontologies. In fact, these ontologies appear largely siloed, especially with regard to sustainability, as developed in different and disconnected contexts, with authors' purposes and goals equally varying for these resources. OBO Foundry-compliant ontologies seem the only exceptions. Lack of integration and interoperability strongly prevents an explicit, fair, and shared enumeration and practice of stakeholder sustainability perspectives, contrasting effective understanding of sustainability.

5 | Considerations and Conclusions

We have presented major theoretical challenges to sustainability and we assessed available ontologies accordingly. We now dedicate this section to outlining some considerations on our approach and to offering possible future directions of work.

5.1 | Possible Advancements

Our assessment relied significantly on manual procedures, making it prone to errors in retrieving and cataloging data. Its time-consuming nature limited the number of resources that could be assessed too. Data sorting may equally be affected by personal biases on the themes investigated. We envisage different strategies to improve our approach. For instance, Artificial Intelligence (AI) tools, such as Large Language Models (LLMs) and Natural Language Processing (NLP), may be employed to semi-automate and enhance the data-gathering and assessment processes (for an example, see the work of Lange et al. 2025). Manual verification will follow. Implementations of our approach will strengthen the analysis of sustainability ontologies.

We documented the several resources evaluated in a List of Sustainability Ontologies. We envision advancing it into an open-source Ontology Catalog to establish a common reference where consulting, retrieving, storing, and contributing to ontologies and terminology for sustainability. This catalog will cover supplementary information as well as additional resources and terminology addressing sustainability and related fundamental domains, such as for instance, health and food. The Ontology Catalog will foster participatory and interoperable construction and alignment of sustainability ontologies and related applications.

5.2 | Toward a Family of Sustainability Ontologies

The shortcomings highlighted in our ontology assessment recommend building a family of interoperable ontologies for sustainability studies and initiatives. We aim to set up a reference in ontology to support and promote coordination and integration of sustainability research and intervention efforts across fields of study, stakeholders, and communities.

We opt for a family of interoperable resources over a single, over-comprehensive ontology for the following reasons: (1) a family of interconnected ontologies would be more likely to ensure efficacy in capturing, clarifying, and integrating nuances belonging to sustainability across perspectives, domains, and approaches; a single ontology would hardly be able to do so in a consistent fashion, limited by conceptual and practical design reasons; (2) the development and maintenance of a single ontology would also require too much time and effort compared to a series of integrated resources, which could capitalize on the work accomplished so far; (3) a family of interconnected ontologies would more efficiently support sustainability communities in dialogue with each other, while commonly implementing integrated approaches. For instance, stakeholders may employ one of the ontologies of the family to implement a dashboard or a platform addressing specific sustainability features according to a given perspective, while others may do the same using another ontology to track a different view. Nevertheless, as employed ontologies are interoperable, a common semantic layer is in place, fostering data integrability across applications toward common actions for sustainability.

Here we offer a potential trajectory to design a family of sustainability ontologies. Future publications may extend the

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considerations we make here. A family of sustainability ontologies requires to be designed based on a foundational structure to strengthen its consistency and integrity concerning sustainability and related perspectives (Figure 1). The foundational structure we envisage comprises BFO as the upper-level ontology to foster representation consistency and access to OBO-Foundry's work; SDGIO as the bridge to the UN frame; and a novel ontology called Sustainability Core Ontology (SCO) (Annex 2) that models sustainability theoretical challenges as the central backbone for integration and coherence with regard to sustainability. Our choice of interfacing with OBO-Foundry ontologies is motivated by OBO Foundry's strong commitment to interoperability, and by the capital role OBO Foundry plays in the field of applied ontology. We envision investigating and interconnecting with other communities and approaches to ontology in the future. For the sake of brevity, this paper does not detail SCO. See Ubbiali (2025) for an analysis of this ontology, and follow this link to consult its most current version and its development protocol. We envision harmonizing available ontologies with the suggested foundational structure.

Despite not completely addressing the challenging nature of sustainability, the Meta-Model ontologies and FEWsOnto appear as interesting candidates from which to start harmonizing ontologies. As available sustainability ontologies may employ some existent upper-level ontologies, different from BFO, mapping them to the foundational structure appears as a further, necessary step for fostering the harmonization of these resources within the family. Finally, we envisage designing, based on the foundational structure, new ontologies that will model sustainability and sustainable development within fields of knowledge for which an ontological resource is not available yet, for example, a food systems sustainability ontology. See the Sustainable Meat Systems Ontology (SuMSO) for an example of a novel sustainability ontology integrating this family (Ubbiali 2025). Establishing a cluster of related (meta-)frameworks, each delineating foundational elements of sustainability within different domains of interest, is key to support a consistent development of this ontology family to advance sustainability efforts.

5.3 | A Family of Sustainability Ontologies in Action

The interconnected ontological resources for sustainability described in Subsection 5.2 will empower sustainability efforts in several ways. Elucidating meanings and making terminologies corresponding across perspectives and approaches, this family of ontologies may support designing consistent correspondences among alternative corpora of metrics and indicators, across continents, nations, regions, and communities. For instance, indexes, such as the Canadian National Index on Agri-Food Performance and the EU food system monitoring framework, may be mapped to each other or to common international standards to establish shared, capillary, and interconnected transoceanic monitoring practices of sustainability efforts. Similarly, sustainability scholars and practitioners may employ family ontologies to align sustainability theories and frameworks of reference (see, for instance, the https://scientificadvice.eu/advice/a-sustainable-food-system-for-the-european-union/"sustainable food system for the European Union" EU report (European Commission et al. 2020)), endorsed by different stakeholders across different fields of study, supporting cross-domain and viable interpretations of sustainability and sustainable development. Further, family ontologies can be implemented into digital applications, such as interactive dashboards and platforms (see, for instance, the Food System Dashboard and k.LAB from the ARIES project)

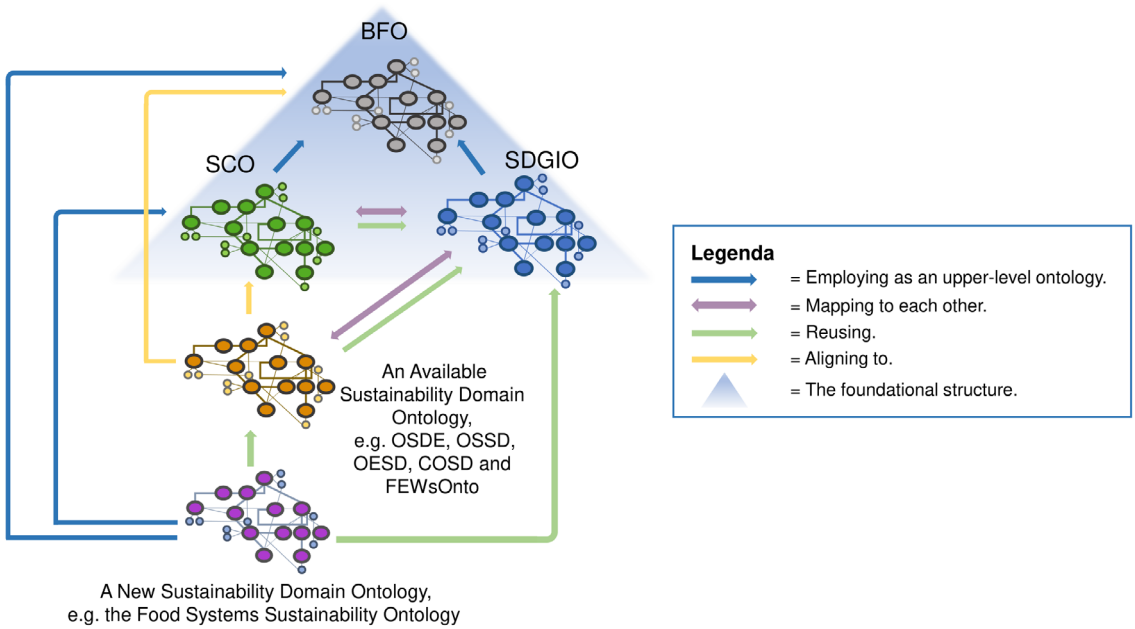


FIGURE 1 | Graphical representation of the family of sustainability ontologies. The light-blue triangle identifies the foundational structure, composed of Basic Formal Ontology (BFO) (grey), Sustainability Core Ontology (SCO) (green), and Sustainable Development Goals Interface Ontology (SDGIO) (blue). Arrows denote the harmonization of pre-existent (yellow) and novel (purple) ontologies with the foundational structure. This figure provides the legenda for arrow colours and examples of ontologies to harmonize.

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to supply semantically sound layers in support of data annotation, computation, and interoperability. Such applications will benefit from interconnected vocabularies and semantics that will strengthen linkage and interoperation, ultimately fostering integrated monitoring, analysis, and interventions across projects and fields of study. Finally, more broadly, this family of sustainability ontologies may be the first step toward establishing a Community of Common Ontologies, Semantics, and MOdels for Sustainability (COSMOS) that will pave the way toward synergically integrating and fostering participatory initiatives and efforts for sustainability across disciplines, stakeholders, and communities.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Annex 1 provides the List of Sustainability Ontologies, and includes three spreadsheets: (1) the Annex guidelines, (2) the list of assessed ontologies (List A), and (3) the list of assessed terms (List B). It is available at the following link: https://github.com/gioUbbiali/The-List-of-Sustainability-Ontologies. Annex 2 provides the current version of SCO and its initial Structured Vocabulary; it includes two spreadsheets, pertaining to the SCO Structured Vocabulary: (1) guidelines, and (2) the tabular and graphical representation of the vocabulary. Both SCO and the Structured Vocabulary are available at the following link: https://w3id.org/sco/repo. Methodologies for annex construction are available within the paper Subsection 4.1, the annexes' guidelines spreadsheet, and the ontology development protocol.

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